B.Sc (HONS.) IN CSE FIRST YEAR FIRST SEMESTER EXAMINATION, 2022

ELECTRICAL AND ELECTRONIC CIRCUIT

[According to the New Syllabus]

Subject Code : CSE-510203

Time—3 hours

Full marks—80

[*N.B. The figures in the right margin indicate full marks. Answer any* **four** *questions.*]

Marks

1. (*a*) Define alternating current and phase. 4

(*b*) Derive the average value of an alternating current. 4

(*c*) What do you mean by resonance of a circuit? Show that the resonance frequency of a series R-L-C circuit is $f_0 = \frac{1}{2\pi\sqrt{LC}}$. 6

(*d*) A coil having an inductance of 50 mH and resistance 10Ω connected in series with a 25μF capacitor across a 200V AC supply. Calculate : 6

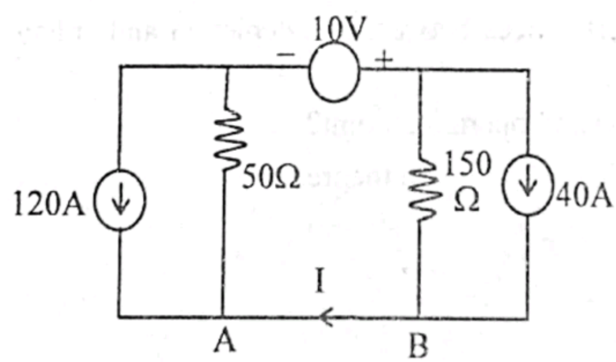
(*i*) Resonance frequency of the circuit

(*ii*) Current flowing at resonance condition

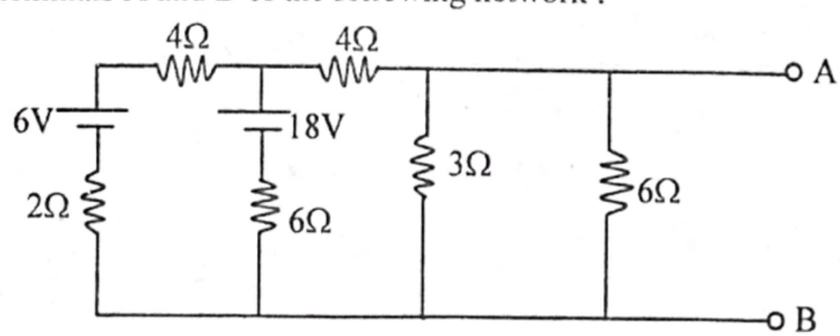
(*iii*) Q-factor.

2. (*a*) State and explain the KCL and KVL. 6

(*b*) Find the current I of the following circuit using super position theorem : 6



(*c*) Obtain the Thevenin's equivalent circuit with respect to the terminals A and B of the following network : 8



[*Please turn over*

Marks

3. (*a*) Explain the classification of material according to energy bands. 5

   (*b*) What is doping? Explain the doping process for extrinsic semiconductors. 5

   (*c*) What is p-n junction? Describe the conditions of a p-n junction when it is under (i) no bias, (ii) forward bias and (iii) reverse bias position. 10

4. (*a*) What is transistor? Describe the working principle of transistor. 6

   (*b*) Describe the operation of transistor as an amplifier. 4

   (*c*) Draw and explain input and output characteristics of a common base configuration of BJT. 6

   (*d*) Distinguish between BJT and FET. 4

5. (*a*) What is operational amplifier? What are salient features of op-Amp? 4

   (*b*) Define CMRR and skew rate. 4

   (*c*) How an operational amplifier can be used as : 6

   (*i*) Differentiator

   (*ii*) Summing amplifier.

   (*d*) What is feedback? How negative feedback reduces noise and non-linear distortion? 6

6. (*a*) Describe the basic operation of JFET with diagram. 6

   (*b*) What are the differences between the depletion and enhancement MOSFET? 4

   (*c*) What is load line and operating point? 4

   (*d*) State and explain superposition theorem. 6

B.Sc (HONS.) IN CSE, FIRST YEAR FIRST SEMESTER EXAMINATION, 2022

ENGLISH

[*According to the new syllabus*]

Subject Code : 510209

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any **four** *of the questions. Each part of a question must be answered sequentially]*

Marks

1. Read the following passage and answer the questions below :

War is the worst of all evils of human civilization. It is an ancient institution which has existed for at least six thousand years. It was always wicked and usually foolish, but in the past, human race managed to live with it. Modern ingenuity has changed this drastically. Either man will abolish war or war will abolish him. In the present, it is nuclear weapons that cause the gravest danger, but bacteriological or chemical weapons may, before long, offer an even greater threat. If we succeed at any cost in abolishing nuclear weapons, our work will not be done completely. It will never be completed until we have succeeded in abolishing war for ever. To do this, we need to persuade mankind to look upon international questions in a new intellectual and mutual way, not as contest of force or power, in which the victory goes to the side which is the most skillful in massacre, but by arbitration in accordance with agreed principles of law. It is not easy to change age-old mental habits, but this is what must be attempted. It is the destructive war that must be checked for the existence of life on earth.

1. (*a*) *(i)* Why is war more dangerous at present? 5

(ii) How should all international problems be solved?

(*b*) What is the main idea and what are the supporting ideas of the passage? 5

(*c*) Write down the meaning of the following words in English and make your own sentences with them (any **five**): 5

Arbitration, drastically, abolish, gravest, persuade, destructive, existence.

(*d*) Write a summary of the above passage. 5

2. (*a*) Frame WH-Questions from the following sentence: 5

(*i*) The news makes him laugh.

(*ii*) She comes to my home once a week.

(*iii*) The lady looked like an actress.

(*iv*) Rajshahi is famous for mangoes.

(*v*) The letter has been written by the boy.

[Please turn over

Marks

(*b*) Use the right form of verbs in the following sentences: 5

(*i*) It is many years since I (come) to Dhaka.

(*ii*) I already (forget) his name.

(*iii*) While I (go) to the market, I met with Mr. Abeg.

(*iv*) He talked as if he (know) everything.

(*v*) If I had a camera, I (take) the snap.

(*c*) Correct the following sentences: 5

(*i*) Take care of your luggages

(*ii*) Twenty dollars a week (not go) far.

(*iii*) My father died by covid-19.

(*iv*) The girl knows to drive a car.

(*v*) Everybody has done their duty.

(*d*) Combine each of the following groups of sentences into one sentence: 5

(*i*) Reading is a good habit, It costs little. It brings immense benefits. The habits should be cultivated.

(*ii*) Self-reliance means self-help. It is a great virtue. Everybody should practice it.

3. (*a*) Amplify the followings: 10

God helps those who help themselves.

(*b*) Translate into English :

দুর্নীতি একটি সামাজিক রোগ। উন্নয়নের পথে এটা বড় বাধা। দুর্নীতি সমাজকে ধ্বংস করে। সমাজের উন্নয়নের জন্য আমাদের দুর্নীতি প্রতিরোধ করা অপরিহার্য। আমাদের সবার উচিত দুর্নীতিকে ঘৃণা করা।

4. Write an essay on any **one** of the followings: 20

(*a*) E-Learning.

(*b*) Environment Pollution and Global Warming.

(*c*) Student and Social Service.

(*d*) Empowerment of Women in Bangladesh.

5. (a) Write a dialogue between two friends about the Metro Rail in Bangladesh. 10

(*b*) Write a report on a newspaper on the celebration of "Pahela Falgun" on your college campus. 10

6. Write a paragraph on any **one** of the followings: 10

(*a*) (*i*) Smart Bangladesh.

(*ii*) Self-Employment.

(*b*) Write an application to the editor of a newspaper criticizing the adoption of unfair means in the examination hall. 10

B.Sc (HONS.) IN CSE FIRST YEAR FIRST SEMESTER EXAMINATION, 2022

PHYSICS

[According to the new syllabus]

Subject Code : CSE 510207

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any **four** *questions.]*

Marks

1. *(a)* Define charge density and electric field intensity. 2

   (b) Derive the relation between surface charge density (δ) and electric field intensity (E). 5

   (c) What is electric field intensity? Deduce an expression for torque (τ) and potential energy (u) by an electric dipole placed in a uniform electric field. 2+8=10

   (d) What do you mean by conservation of charge? 3

2. *(a)* State and explain Coulomb's law. 3

   (b) What is electric dipole? Derive an expression relating to there electric vectors. 1+5=6

   (c) Define potential and deduce the relation between potential and field strength. 5

   (d) What is electric dipole? Find the electric field strength of a dipole at any point along the dipole axis. 6

3. *(a)* What is capacitance? Calculate the capacitance of a parallel plate capacitor. 6

   (b) Derive the expression of the equivalent capacitance for a series combination of capacitors. 4

   (c) Define conductance and conductivity. 2

   (d) Describe the construction of a galvanometer and describe it's operation. 8

[Please turn over

Marks

4. *(a)* Define magnetic field. Calculate magnetic field along the axis of circular coil. 6

   (b) Show that magnetic force acting on a current carrying wire of length L is $\vec{F} = i\,(\vec{L} \times \vec{B})$ where the symbols have conventional meaning. 6.

   (c) Explain Fleming's left and right hand rule. 4

   (d) State and explain Ampere's law. 4

5. *(a)* What is Lorentz force? Explain. 5

   (b) Define magnetic flux. Explain Biot-Savart law and express it in vector form. 2+8=10

   (c) Find out self inductance of two parallel wires. 5

6. *(a)* Explain Maxwell's Cork Screw rule. 3

   (b) Derive the charging equation of R—L circuit with appropriate diagram. 6

   (c) Derive the expression for the energy stored in magnetic field. 7

   (d) State and explain Ohm's law. 4

B.Sc (HONS.) IN CSE FIRST YEAR FIRST SEMESTER EXAMINATION, 2022

STRUCTURED PROGRAMMING LANGUAGE

[According to the New Syllabus]

Subject Code : CSE-510201

Time—3 hours

Full marks—80

[*N.B. The figures in the right margin indicate full marks. Answer any* **four** *questions.*]

Marks

1. (*a*) Define structured programming language. Discuss the basic structure of C Programming Language. 2+5=7

(*b*) Define the following terms : 3

(*i*) Algorithm

(*ii*) Flowchart

(*iii*) Pseudo-code.

(*c*) Write down an algorithm and flowchart to calculate the temperature Fahrenheit to Celsius. 5

(*d*) Write a program to calculate area and perimeter of a circle. 5

2. (*a*) What is token? Describe different types of token. 1+4=5

(*b*) What is variable? Differentiate between local and global variable. 1+4=5

(*c*) Describe the four basic data types used in C language. How could we extend the range of values they represent? 5

(*d*) Write a program solution of quadratic equation ($ax^2 + bx + c = 0$). 5

3. (*a*) What are the differences between $++y$ and $y++$? 4

(*b*) Describe the following statements with general form and flowchart : 6

(*i*) if....else

(*ii*) for

(*iii*) switch

(*c*) Write a C program that prints the smallest number from three numbers. 5

(*d*) Consider the following C code and findout every value of the variables (x, y, z)— 5

```
int x = 5, y = 8, z = 3 ;
x = x + (z + +) ;
printf (" %d %d", x%y, x/y) ;
y = − − x ;
printf(" %d %d %d", x, y, z) ;
```

[Please turn over

Marks

4. (*a*) Differentiate with syntax between structure and union. 4

   (*b*) What is string? Write down four string handling functions with their actions. 1+4=5

   (*c*) What is storage class? Describe different types of storage class in C program. 6

   (*d*) Write a program that check whether a string is palindrome or not. 5

5. (*a*) What is an array? Write down the advantages and disadvantages of array. 1+4=5

   (*b*) Discuss different types of operator used in C Language. 5

   (*c*) What do you mean by function in C program? Write down the differences between actual and formal parameters. 1+4=5

   (*d*) Define pointer. Write down a C program to exchange the value of two variables (x, y) using pointer. 1+4=5

6. (*a*) What is the significance of the EOF used in file? Write down the advantages of file. 5

   (*b*) What is preprocessor directive? Write down the functions of <stdio.h> and <math.h> header files. 1+4=5

   (*c*) Define file handling mode. Discuss modes for opening a file used in C language. 1+4=5

   (*d*) Write down a C program to read the contents of a file and write the contents to another file. 5

Institute of Science and Technology (IST)
CSE 1st Semester Incourse-1 Examination 2023
CSE 510201: Structured Programming Language
Full Marks: 20 Time: 50 minutes

Answer any two of the following Questions

1.	a)	Draw and explain components of a computer.	4
	b)	List out importance of C program.	3
	c)	Write a program to display the text "Hello Students!".	3
2.	a)	Define pseudocode. Write an algorithm and draw a flowchart to find average of three numbers taken as input from the user.	1+3=4
	b)	Differentiatebetween intmain() and void main().	3
	c)	Mention the rules for identifiers.	3
3.	a)	Compare compiler and interpreter.	3
	b)	Describe the basic structure of C program.	5
	c)	What would be the value of x after execution of the following statements? **intx,y=7; char z= 'b'; x=z+y;**	2

Good Luck!

Institute of Science and Technology

In course Examination-2023

B.Sc (Hons.) in CSE, 1st Semester

Calculus (CSE-510205)

Time: 50 minutes **Full Marks: 20**

1. Find $\frac{dy}{dx}$ (any **three**) 4x3=12

   i) $x^y + y^x = a^b$

   ii) $y = (\sin x)^{\ln x} + \sin^2(\cos^{-1} x)$

   iii)

   $$x = 3\tan^{-1}\frac{2t}{1-t^2}, y = 2\sin^{-1}\frac{2t}{1+t^2}$$

   iv)

   $$y = \tan^{-1}\left(\frac{1+\tan x}{1-\tan x}\right)$$

2. State Leibnitzs theorem. If $y = \sin(m \sin^{-1} x)$ then prove that, $(1-x^2)y_{n+2} - (2x+1)xy_{n+1} + (m^2 - n^2)y_n = 0.$ 2+6=8

B.Sc (HONS.) IN CSE FIRST YEAR FIRST SEMESTER EXAMINATION, 2022

CALCULUS

[According to the New Syllabus]

Subject Code : CSE-510205

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any **four** *questions.]*

Marks

1. (*a*) Define algebraic and trigonometric functions. 4

(*b*) Draw the graph of the following function : 8

$$f(x)=\begin{cases} x^2+1 & \text{when} \quad x<0 \\ x & \text{when} \quad 0\le x\le 1 \\ \frac{1}{x} & \text{when} \quad 1<x \end{cases}$$

Find the domain and range of the above function.

(*c*) A function $f(x)$ is defined as follows : 8

$$f(x)=\begin{cases} -x & \text{when} \quad -2\le x\le 0 \\ x & \text{when} \quad 0<x<1 \\ 3-x & \text{when} \quad 1\le x\le 2 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 0$ and $x = 1$.

2. (*a*) Answer any **three** : 4×3=12

Find $\frac{dy}{dx}$ when—

(*i*) $y = \log_a x + \log_x a$

(*ii*) $y = \tan^{-1}\left(\frac{\cos x + \sin x}{\cos x - \sin x}\right)$

(*iii*) $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$

(*iv*) $x = a\cos^3\theta, y = a\sin^3\theta$.

(*b*) State Leibnitz's theorem. *If* $y = \sin(m\sin^{-1}x)$ then prove that, 2+6=8

$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0$

3. (*a*) State and prove Lagrange's mean value theorem. 8

(*b*) Verify mean value theorem for the function $f(x) = 3 + 2x - x^2$ in the interval (0, 1). 6

(*c*) Determine the maximum and minimum values of the function $\sin x + \cos 2x$ for $0 < x < 2\pi$. 6

[Please turn over

Marks

4. (*a*) Evaluate (any **five**) : 4×5=20

(i) $\int \frac{\sin^3 x}{\cos^9 x} dx$

(ii) $\int \sin^{-1}\sqrt{\frac{x}{a+x}}\, dx$

(iii) $\int \frac{dx}{(x+1)\sqrt{x+3}}$

(iv) $\int \frac{(1+x)e^x}{\cos^2(xe^x)} dx$

(v) $\int \sin^5 x \cos^3 x \, dx$

(vi) $\int \frac{dx}{\sqrt[3]{x}\ \sqrt[3]{(1+x)^5}}$

5. (*a*) Evaluate any **two** of the followings : 4×2=8

(*i*) $\int_0^{16} \frac{x^{\frac{1}{4}}}{1+\sqrt{x}} dx$

(*ii*) $\int_0^{\pi} \frac{dx}{a^2-2ab\cos x+b^2}$

(*iii*) $\int_0^{\frac{\pi}{4}} \ln(1+\tan\theta)\, d\theta$

(*b*) State and prove fundamental theorem of definite integral. 6

(*c*) Evaluate $\underset{n\to\infty}{Lt}\ \frac{1}{n}\ [(n+1)(n+2)(n+3)\ \ldots\ldots..(n+n)]^{\frac{1}{n}}$. 6

6. (*a*) Find the length of the perimeter of the cardioide $r = 8(1+\cos\theta)$. 6

(*b*) Find the whole area of the hypocycloid 7

$x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}$.

(*c*) Find the volume of the solid generated by revolving the ellipse 7

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ about x-axis.